

Sophia Huang

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EDUCATION

Binghamton University, State University of New York

Master of Science in Data Analytics – School of Management

Binghamton, NY

July 2026

Cumulative GPA: 3.93/4.00

Relevant Courses: Machine Learning & Data Mining, AI Driven Decision Support, Analytics for Enterprise Apps

Bachelor of Arts in Psychology, Minor in Digital and Data Studies – Harpur College

May 2025

Cumulative GPA: 3.86/4.00 | **Dean's List Spring 2022 – Spring 2026**

Instituto Lorenzo de' Medici

Florence, Italy

Italian Study Abroad Program, Gilman Scholarship Recipient

May 2024

TECHNICAL SKILLS

Programming & Data Analysis: Python, R, SQL, Bash

Libraries & Tools: PyTorch, Pandas, NumPy, TensorFlow, Scikit-learn, Fastai, Shiny, AWS (Lambda, EC2, DynamoDB, S3), Linux, Figma, Redshift, Neo4j, MongoDB

Visualization & Analysis: Tableau, Power BI, Excel

Statistical Concepts: Hypothesis Testing, Regression Analysis, ANOVA, Probability Distribution, Time Series Analysis

Other: 2nd Degree Black Belt in Karate, Mandarin (Fluent)

PROFESSIONAL EXPERIENCE

Cognition, Aging, and Memory Performance (CAMP) Lab, Binghamton University, Vestal, NY

Research Data Scientist

October 2025 – May 2026

- Engineered an automated Python pipeline using pandas and NumPy to transform raw eye tracking data files into analysis-ready data for 100+ participants, reducing errors by 90% and processing time from 3-4 hours to seconds
- Developed entropy metric calculations by implementing reusable functions applied across fields and participants, producing comparable features for statistical modeling and reporting on 400+ research variables

Information Technology Services, Binghamton University, Vestal, NY

Lead Student Help Desk Analyst

August 2023 – May 2026

- Promoted to lead analyst after 2 years, supervising and onboarding 11 student workers and coordinating schedules to ensure seamless workflow and service delivery across a team supporting 100+ users
- Achieved a 90% first-contact resolution rate across 300+ annual support tickets by implementing streamlined troubleshooting protocols, minimizing escalations and optimizing support efficiency
- Diagnosed and resolved software and hardware failures for 50+ Mac and Windows machines, applying root cause analysis to restore functionality for screens, keyboards, and system errors
- Received positive feedback from over 95% of campus community members supported via phone, email, and in-person interactions, translating complex technical concepts into clear, accessible explanations

PROJECT EXPERIENCE

JPMorgan Chase, Planning vs. Actual Usage Analysis

New York, NY

Group Leader

May 2026 – July 2026

- Analyzed 127,000+ infrastructure cost records, building OLS regression models and diagnostic visualizations in Python to quantify a systematic 4x overestimation in planned versus actual spending across 672 applications
- Delivered data-driven recommendations, including volume-based forecasting, organizational hierarchy simplification, and monthly variance tracking, presented in a polished executive slide deck
- Automated summary statistics, Excel reporting, and chart generation across 13 cost dimensions using pandas, openpyxl, and matplotlib

Binghamton University, Campus Parking Availability Detection

Vestal, NY

Team Member

January 2026 – May 2026

- Built a computer vision pipeline using YOLOv2 to detect and classify parking space occupancy from lot images, archiving 90% precision and 93% recall on a 6,000+ image test set of 255,000+ labeled spaces
- Designed an interactive mobile application prototyped in Figma featuring real-time color-coded lot maps, section breakdowns, peak hours charts, and day/time filtering to help commuters make informed parking decisions
- Established a full scalability framework for deploying the model on Binghamton University's existing camera infrastructure with minimal hardware investment

INTERESTS

Photography - Karate - Boxing - Rock Climbing - Snowboarding - Traveling